

KOBE, Japan

WMO: 477700

Lat: 34.6969N

Lon: 135.2128E

Elev: 100

StdP: 14.64

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	32.3	34.2	16.7	12.8	38.0	19.5	14.7	39.2	24.5	42.4	21.7	43.2	9.6	280	0.432

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.2	92.1	77.5	90.1	77.1	88.3	76.6	79.8	87.1	79.1	86.1	78.4	85.3	9.8	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.9	145.4	83.9	77.1	141.5	83.6	76.3	137.8	83.3	43.5	87.3	42.7	86.3	42.0	85.4	84.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
22.2	18.9	16.4	DB	28.7	95.7	2.9	2.1	26.7	97.2	25.0	98.5	23.4	99.7	21.3	101.2
			WB	25.2	81.0	2.6	1.2	23.3	81.9	21.7	82.6	20.3	83.2	18.3	84.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	62.9	42.8	43.7	49.4	59.0	67.9	74.3	81.5	84.2	77.9	67.9	57.2	47.6
	DBStd	15.01	4.10	5.01	5.57	5.72	4.12	3.56	3.85	3.06	4.76	4.93	5.52	5.06
	HDD50	612	225	186	82	6	0	0	0	0	0	0	6	107
	HDD65	2786	688	596	485	191	20	0	0	0	0	26	240	540
	CDD50	5323	3	11	62	275	556	728	978	1062	838	556	221	33
	CDD65	2018	0	0	0	10	110	278	513	597	388	117	5	0
	CDH74	17501	0	0	0	12	247	1303	5262	7103	3242	331	1	0
	CDH80	5809	0	0	0	0	18	175	1766	2960	868	22	0	0
	WSAvg	7.7	8.1	7.6	7.7	7.6	7.5	7.4	7.2	7.7	7.9	7.8	7.4	8.1
	PrecAvg	51.8	1.7	2.5	3.9	4.0	5.6	7.0	6.4	4.1	6.5	4.7	3.2	2.1
(15) Precipitation	PrecMax	67.7	3.9	5.8	6.9	7.3	14.6	15.0	16.9	15.2	16.6	12.5	7.6	4.1
	PrecMin	26.5	0.2	0.6	1.6	1.1	1.3	2.2	1.0	0.6	1.6	1.3	0.7	0.2
	PrecStd	11.1	0.9	1.4	1.3	1.6	3.0	3.0	4.3	3.5	3.5	2.7	1.7	1.2
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.8	62.3	66.6	75.5	82.2	86.8	94.4	95.4	91.1	82.6	72.3	63.7
		MCWB	52.1	57.1	56.3	60.4	65.1	72.9	78.4	77.8	76.3	71.7	62.7	57.7
	2%	DB	53.7	57.9	63.7	72.0	78.9	83.9	91.2	93.0	88.5	79.6	69.7	59.9
		MCWB	47.0	51.3	55.1	59.8	64.3	71.8	77.4	77.7	75.7	69.5	61.7	53.1
	5%	DB	51.5	54.6	61.3	69.8	76.7	81.9	89.1	91.1	86.6	77.6	67.9	57.5
		MCWB	45.2	48.4	53.7	59.3	63.8	71.0	77.0	77.4	75.0	67.8	60.3	50.6
	10%	DB	49.5	51.9	58.9	67.5	74.7	80.1	87.1	89.4	84.7	75.6	65.7	55.4
		MCWB	43.3	45.8	51.6	58.5	63.4	70.2	76.6	77.0	74.3	65.9	58.2	48.9
	0.4%	WB	53.9	58.1	60.7	67.1	71.1	77.2	80.5	80.9	79.8	75.2	66.7	59.6
		MCDB	57.1	61.3	63.2	70.2	74.2	82.6	89.2	88.5	86.1	79.2	69.1	62.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	49.0	53.0	57.4	64.1	69.4	75.8	79.2	80.0	78.3	72.0	63.7	54.7
		MCDB	52.2	56.7	61.7	68.1	73.5	80.4	86.9	87.3	84.8	77.7	68.1	58.5
	5%	WB	46.2	49.6	54.9	62.1	67.7	74.6	78.5	79.5	77.3	69.6	61.6	51.8
		MCDB	50.5	53.3	60.0	67.3	72.5	79.0	85.7	86.6	83.8	75.6	66.3	56.6
	10%	WB	43.8	46.4	52.5	60.0	66.0	73.0	77.9	78.9	76.3	67.4	59.3	49.3
		MCDB	49.0	51.5	58.1	66.0	72.0	77.6	84.7	85.9	82.9	73.9	64.9	54.6
(35) Mean Daily Temperature Range	5% DB	MDBR	10.1	10.9	12.2	12.6	12.1	10.3	9.5	10.2	10.3	10.7	10.8	10.2
		MCDBR	11.9	13.3	14.1	14.2	14.3	12.2	11.7	12.1	11.4	11.8	11.6	11.7
		MCWBR	9.2	10.6	10.8	8.3	7.6	5.7	4.4	4.7	5.7	7.5	8.8	9.0
	5% WB	MCDBR	11.2	11.9	12.6	11.5	10.5	8.6	10.2	10.1	9.6	10.0	10.5	10.9
		MCWBR	10.2	11.3	12.1	9.3	8.2	4.8	4.2	4.3	5.4	8.2	9.7	10.2
(40) Clear-Sky Solar Irradiance	taub		0.415	0.459	0.525	0.557	0.562	0.583	0.585	0.552	0.503	0.461	0.432	0.409
	taud		2.101	2.000	1.857	1.832	1.884	1.904	1.949	2.026	2.127	2.160	2.141	2.130
	Ebn at Noon		240	243	237	237	238	233	231	236	241	240	233	233
	Edh at Noon		40	49	61	66	63	62	59	54	46	41	38	36
(44) All-Sky Solar Radiation	RadAvg		711	912	1230	1509	1651	1496	1584	1627	1274	1032	776	643
	RadStd		50	74	95	132	166	139	208	151	110	101	64	42

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	+1.17	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (36 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page